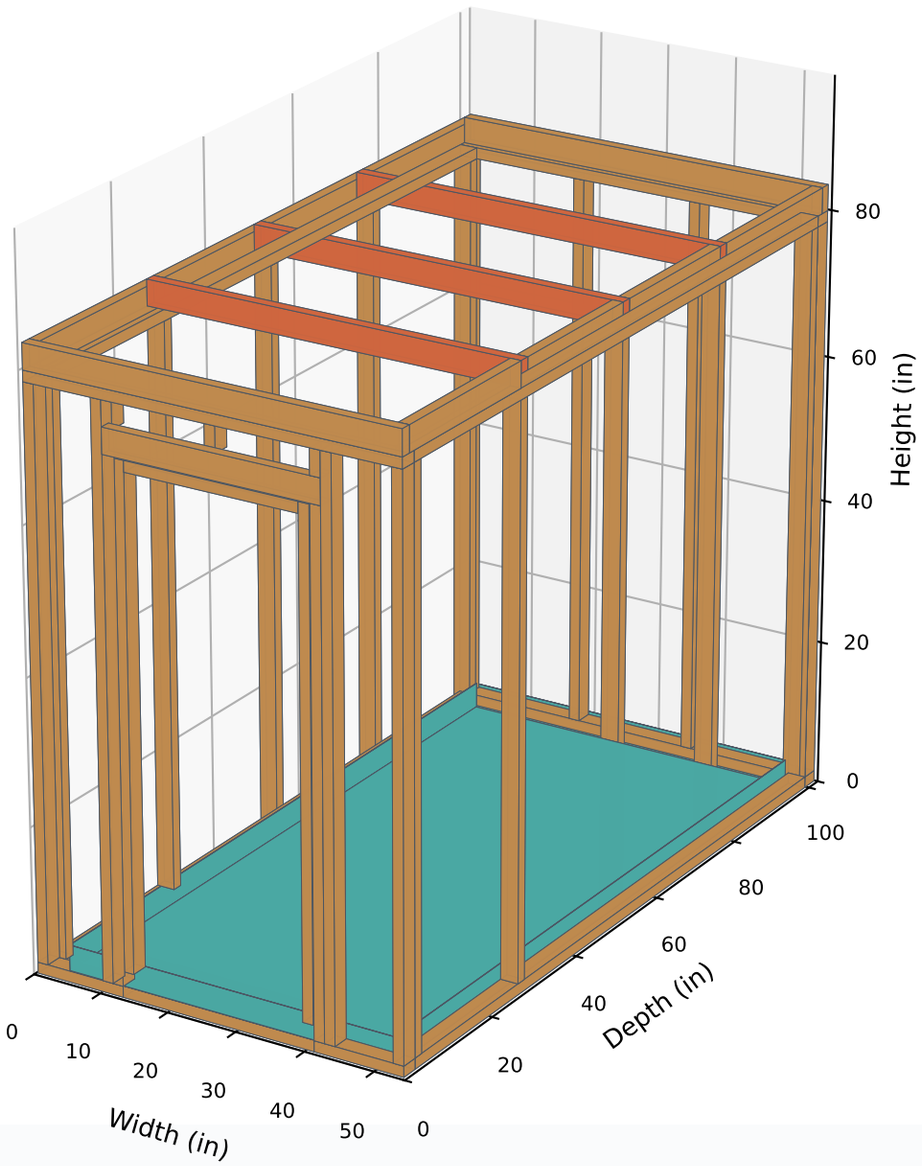
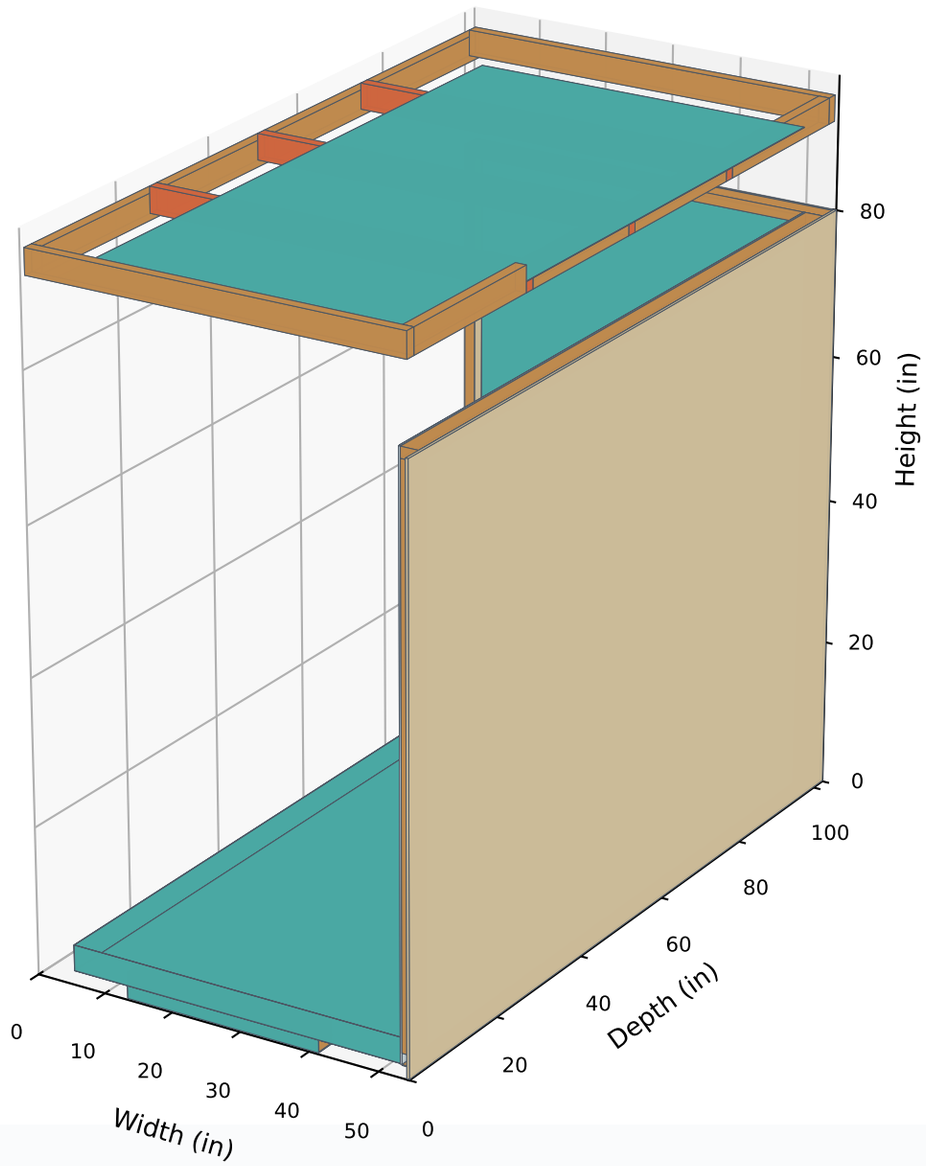
GARAGE GROW LAB / 48 × 96 in interior

Framing with door blank



Cutaway • lid lifted 12 in

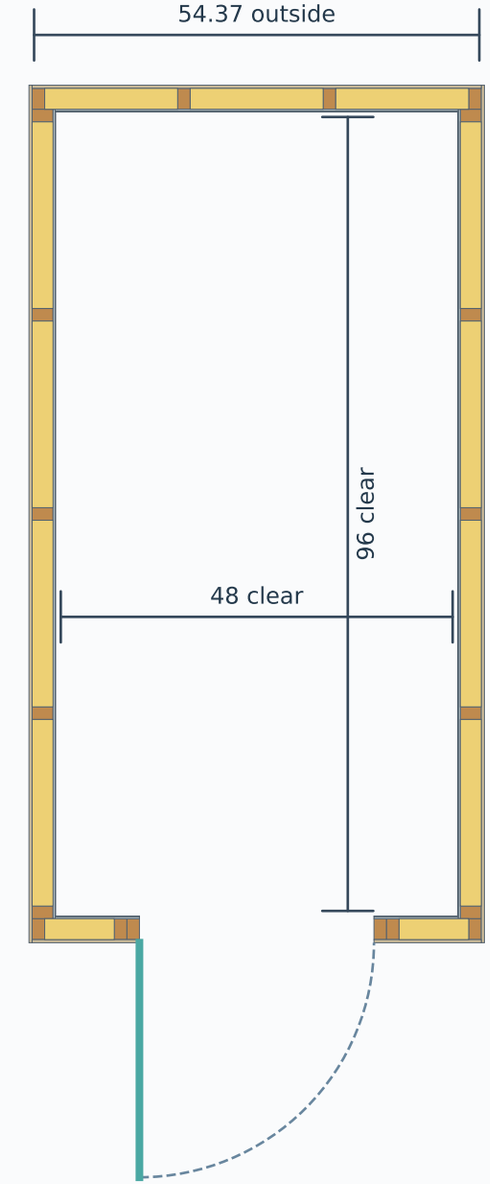


Lumber Plywood Liner Light-support crossmembers

Outside 54.37 × 102.37 × 84 in | Clear height 79.40 in | Slab-supported walls; folded liner floor; no slab anchors

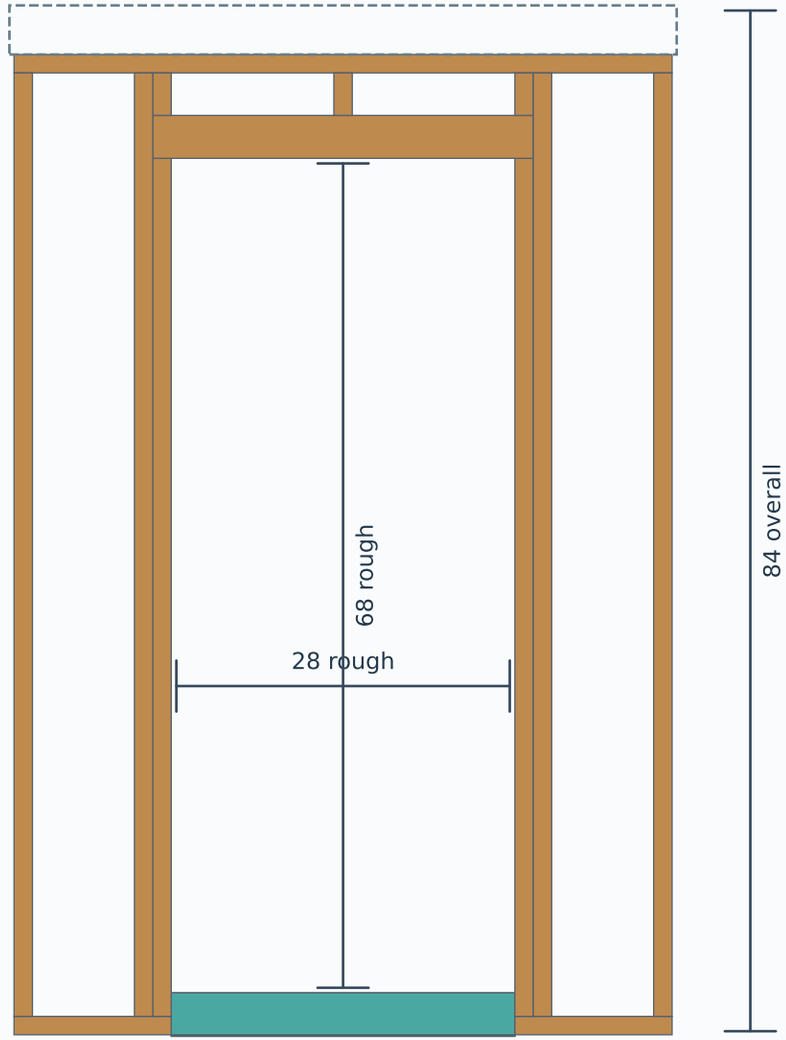
PLAN + FRONT FRAMING / dimensions in inches

Plan at mid-wall; front at bottom



Allow outward door swing + working access

Front panel; door leaf hidden

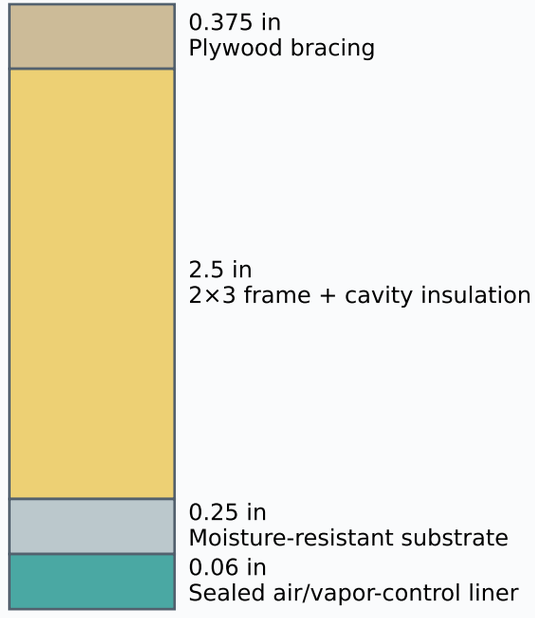


Curb top: 3.54 above slab
Liner floor on concrete; no platform

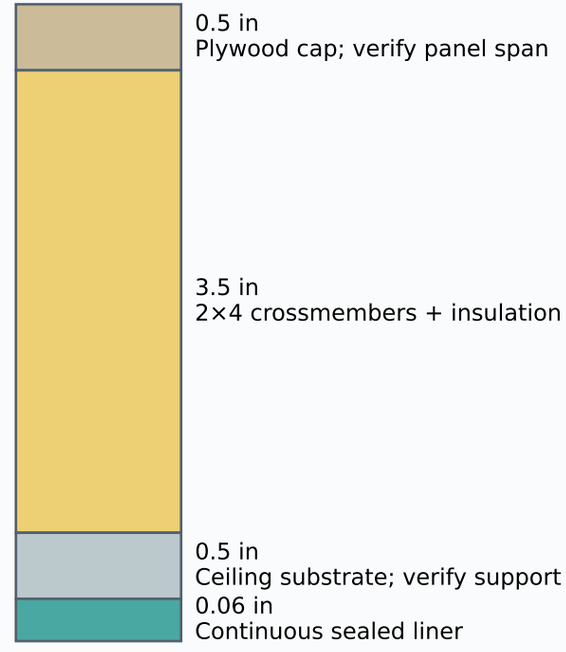
Assembly dimensions, not a fastening schedule. Door jamb/hinge detail must be finalized before cutting the door blank.

ENCLOSURE LAYERS / continuity at every joint

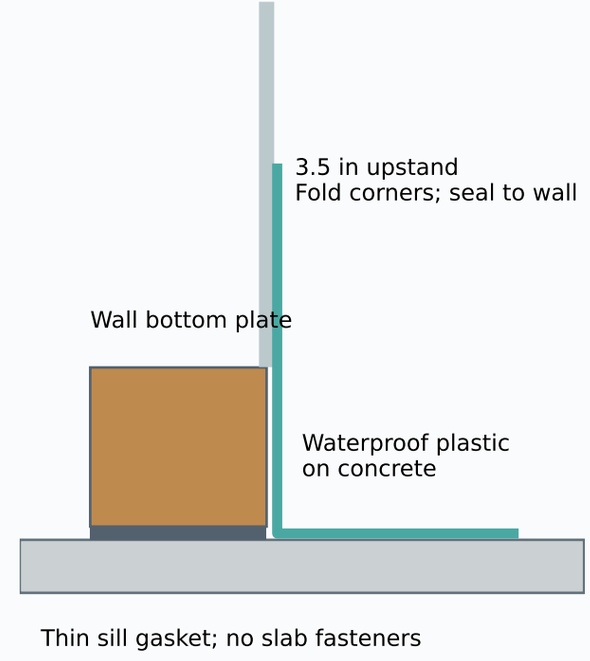
Wall: garage → grow space



Roof: top → grow space



Floor-to-wall fold (schematic)

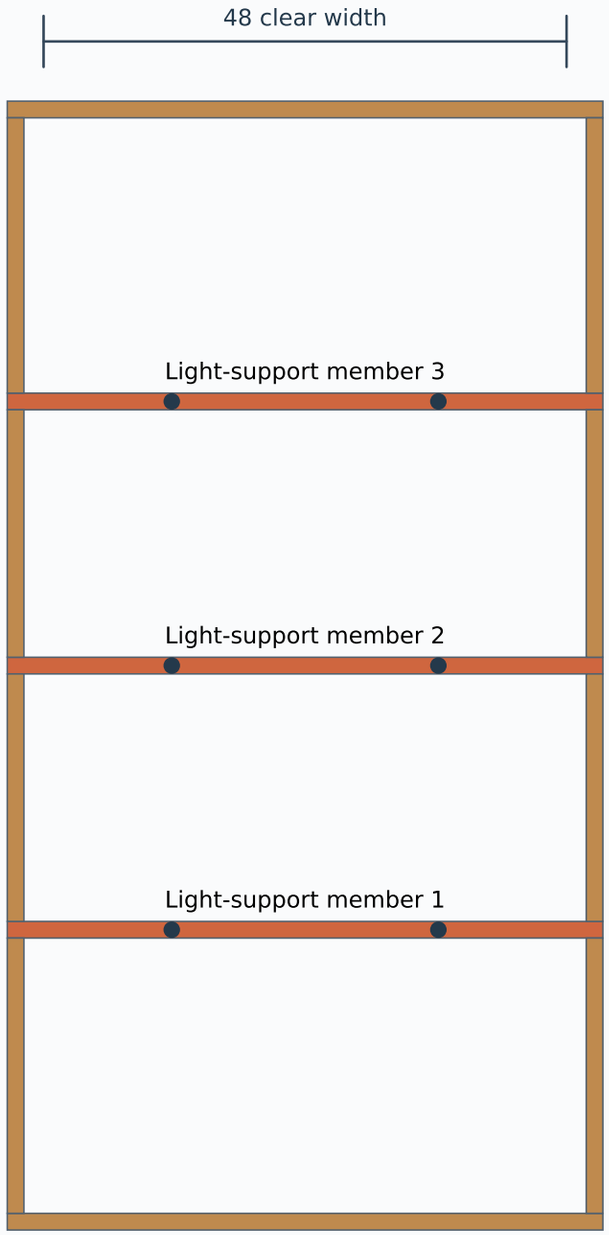


Seal wall-to-ceiling transitions, corner seams, door stops and every hanging/service penetration.
Use one continuous floor liner with folded corners; wrap it over the supported doorway curb.
Keep attachments above the tray rim; protect the liner beneath equipment feet.
Wall plates rest on a thin capillary-break gasket on the slab. No raised floor or floor insulation.

Layer thicknesses labeled; thin layers exaggerated for readability. Cavity R values are not whole-wall R values.
Orange roof members are intended light supports; weights, connections and material capacities remain unverified.

ROOF + LIGHT SUSPENSION / short-span crossmembers

Top view; dots indicate example hanger locations



3 interior 2×4 crossmembers + 2 end members
Actual crossmember section: 1.5 × 3.5 in, on edge
Crossmember cut length: 53.62 in
Maximum roof support spacing: 26.28 in

Load path:
Lights → rated hanging hardware → crossmember
→ wall top plate → aligned side-wall stud → slab

Crossmembers bear directly on wall top plates.
Ceiling finish fits between walls below the framing.
Use hangers fixed into structural wood, with sealed
penetrations through the ceiling air/vapor liner.

Verify roof and ceiling panel span/support requirements;
add seam backing or light furring where required.
Do not hang loads from panel skins or insulation.

Total hanging load: not supplied
This layout is not yet load-rated. See summary.json
for the editable beam-demand study and its limits.